# Colgate Men's Rowing Performance Data

## 1 Overview

This is some work I did for the Colgate rowing team. This is a good example of what could be a test question, we walk through data cleaning, summary, analysis, and modelling.

```
default_chunk_hook <- knitr::knit_hooks$get("chunk")

latex_font_size <- c(
  "Huge",
  "huge",
  "LARGE",
  "Large",
  "large",
  "normalsize",
  "small",
  "footnotesize",
  "scriptsize",
  "tiny"
)

knitr::knit_hooks$set(chunk = function(x, options) {
  x <- default_chunk_hook(x, options)
  if (options$size %in% latex_font_size) {
    paste0("\n \\", options$size, "\n\n", x, "\n\n \\normalsize")
  } else {
    x
  }
})
```

## 2 Charts and stuff

Data cleaning! Remove rowers that quit, and make sure that people that vary their usage of their first and last names are coerced to the same person.

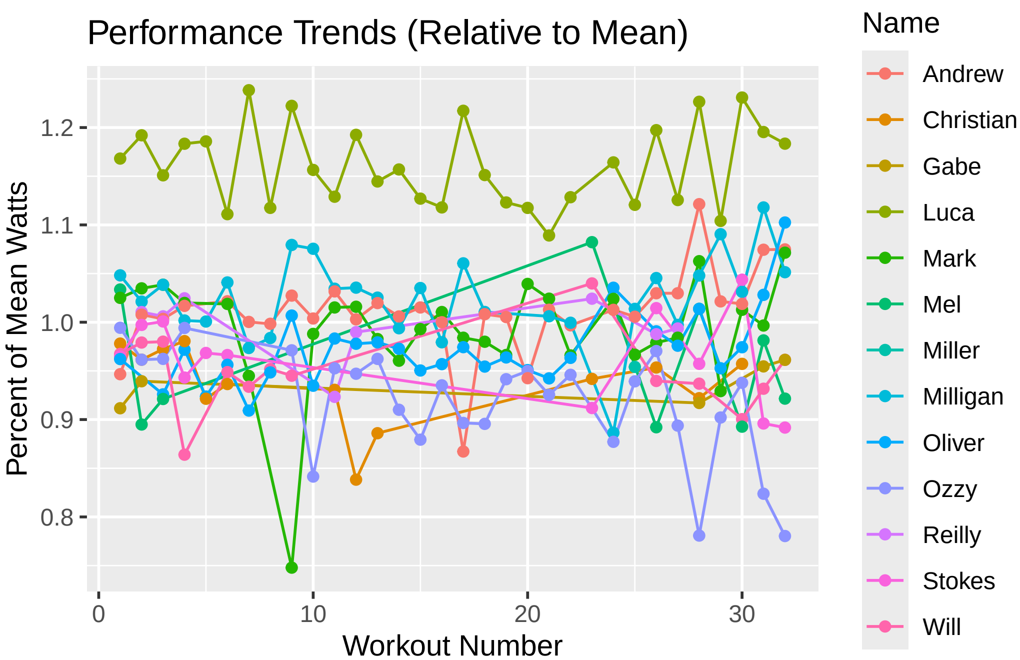
```
data = read.csv("/Users/hudson/Downloads/CSV_Colgate_MROW_Updated_3:12:26.csv") |>
  filter(Name != "Harry" & Name != "David" & Name != "Ethan") |>
  mutate(Name = case_when(Name == "Williams" ~ "Luca",
  .default = Name))
```

```
data = data |>
  group_by(Workout.Number) |>
  mutate('Percent of Mean' = Watts/mean(Watts)) |>
  ungroup()
```

`ggplot`! Try and interpret this plot. How could I have made it better? (Watts are how we measure power in rowing)

```
library(dplyr)
library(ggplot2)
```

```r
library(stringr)

data |>
  mutate(Name = str_extract(Name, "^[A-Za-z]+")) |>
  ggplot(aes(x = Workout.Number,
             y = `Percent of Mean`,
             color = Name,
             group = Name)) +
  geom_line() +
  geom_point() +
  labs(
    x = "Workout Number",
    y = "Percent of Mean Watts",
    title = "Performance Trends (Relative to Mean)"
  )
```



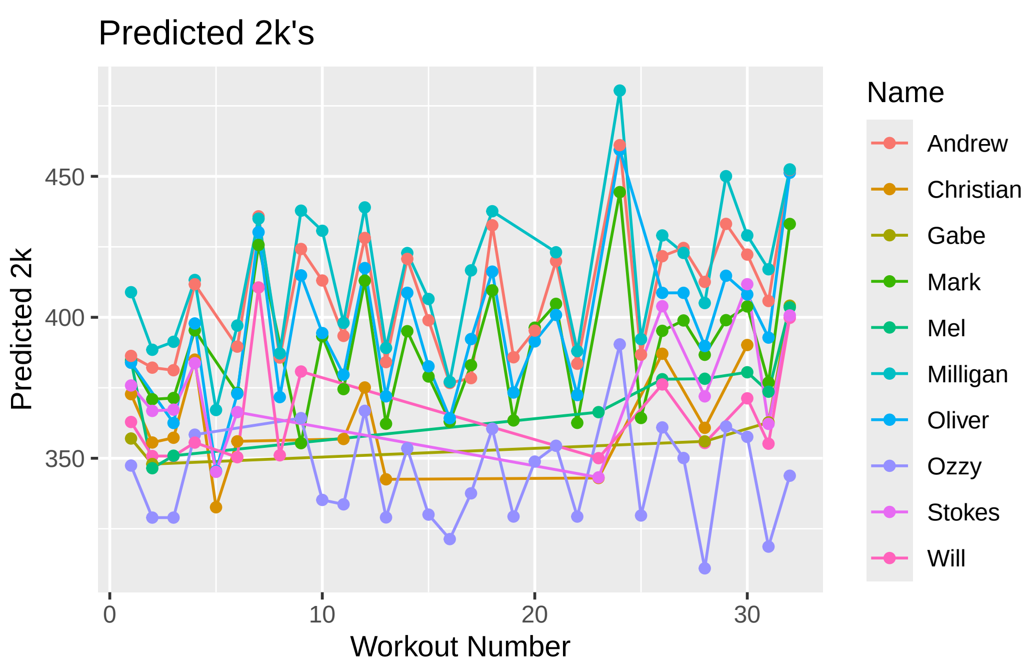

Some more data cleaning! This is converting seconds to watts from the rowers' best races this season.

```r
data <- data |>
  mutate(Name_key = str_extract(Name, "^[A-Za-z]+"),
         Best_2k = case_when(
           Name_key == "Williams" ~ 2.80 * (500 / (365.1 / 4))^3,
           Name_key == "Milligan" ~ 2.80 * (500 / (373.03 / 4))^3,
           Name_key == "Andrew"   ~ 2.80 * (500 / (375.9 / 4))^3,
           Name_key == "Oliver"   ~ 2.80 * (500 / (379.7 / 4))^3,
           Name_key == "Mark"     ~ 2.80 * (500 / (387.6 / 4))^3,
           Name_key == "Mel"      ~ 2.80 * (500 / (388.2 / 4))^3,
           Name_key == "Christian"~ 2.80 * (500 / (390.1 / 4))^3,
           Name_key == "Gabe"     ~ 2.80 * (500 / (393.3 / 4))^3,
           Name_key == "Will"     ~ 2.80 * (500 / (396.4 / 4))^3,
           Name_key == "Stokes"   ~ 2.80 * (500 / (386.0 / 4))^3,
           Name_key == "Ozzy"     ~ 2.80 * (500 / (413.0 / 4))^3,
           Name_key == "Harry"    ~ 2.80 * (500 / (410.7 / 4))^3,
           .default = NA_real_
         )) |>
```

```r
  select(-Name_key)
```

```r
data <- data |>
  mutate(Predicted.2k.new = (Watts * ((Time..s.)^0.19) * Best_2k)^0.455)
```

```r
library(dplyr)
library(ggplot2)
library(stringr)

data |> filter(!is.na(Predicted.2k.new)) |>
  mutate(Name = str_extract(Name, "^[A-Za-z]+")) |>
  ggplot(aes(x = Workout.Number,
             y = `Predicted.2k.new`,
             color = Name,
             group = Name)) +
  geom_line() +
  geom_point() +
  labs(
    x = "Workout Number",
    y = "Predicted 2k",
    title = "Predicted 2k's"
  )
```




This predictive model does not include any numerical summary. If you were to include a numerical summary, what would you report?

```r
data = read.csv("/Users/hudson/Downloads/CSV_Colgate_MROW_Updated_3:12:26.csv")
```

```r
data = data |>
  group_by(Workout.Number) |>
  mutate('Percent of Mean' = Watts/mean(Watts)) |>
  ungroup()
```

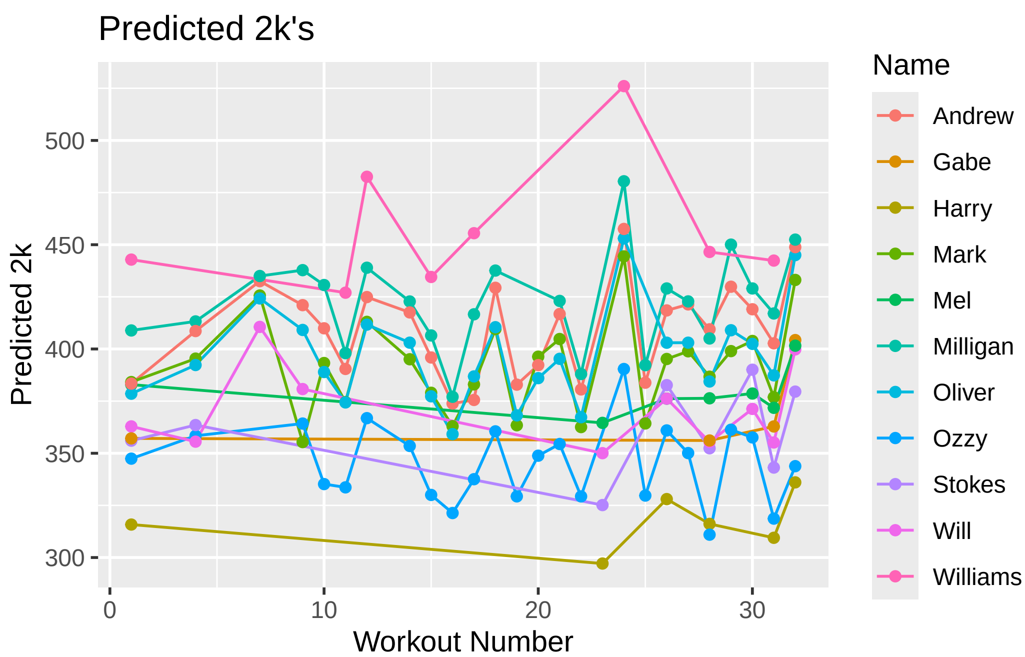
```r
data.no.hr <- data |>
  filter(Time..s. != 3600)
```

```r
data.no.hr = data.no.hr |>
  group_by(Name) |>
  mutate(
    `2k.Watts` = Watts[Workout.Number == 31][1],
    Predicted.2k.new = (Watts * ((Time..s.)^0.19) * `2k.Watts`)^0.455
  ) |>
  ungroup()
```

```r
data.no.hr = data.no.hr |>
    mutate(`Predicted.2k.time.s` = ((2.8 / `Predicted.2k.new`)^(1/3)) * 2000,
    `Predicted.2k.time.m` = sprintf(
      "%d:%02d",
      floor(`Predicted.2k.time.s` / 60),
      round(`Predicted.2k.time.s` %% 60)
    )
  )
```

This is a plot of race predictions under a different model! How would you interpret, change, and summarize this plot?

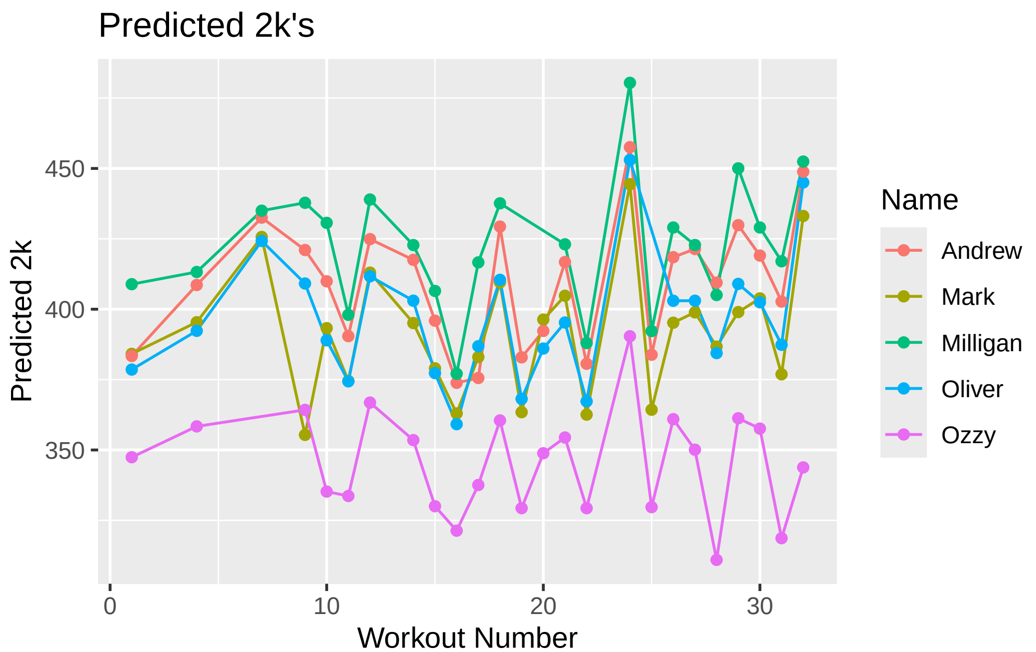
```r
library(dplyr)
library(ggplot2)
library(stringr)

data.no.hr |> filter(!is.na(Predicted.2k.new)) |>
  mutate(Name = str_extract(Name, "^[A-Za-z]+")) |>
  ggplot(aes(x = Workout.Number,
             y = `Predicted.2k.new`,
             color = Name,
             group = Name)) +
  geom_line() +
  geom_point() +
  labs(
    x = "Workout Number",
    y = "Predicted 2k",
    title = "Predicted 2k's"
  )
```




```r
# #| echo: false
# library(dplyr)
# library(ggplot2)
# library(stringr)

# data.no.hr |> filter(!is.na(Predicted.2k.new)) |>
#   mutate(Name = str_extract(Name, "^[A-Za-z]+")) |>
#   ggplot(aes(x = Workout.Number,
#              y = `Predicted.2k.time.m`,
#              color = Name,
#              group = Name)) +
#   geom_line() +
#   geom_point() +
#   labs(
#     x = "Workout Number",
#     y = "Predicted 2k Time",
#     title = "Predicted 2k's (time)"
#   )
```

```r
athletes <- c("Milligan", "Ozzy", "Andrew", "Oliver", "Mark", "Lucca", "Matheson")


data.no.hr |> filter(!is.na(Predicted.2k.new)) |>
    filter(Name %in% athletes) |>
  mutate(Name = str_extract(Name, "^[A-Za-z]+")) |>
  ggplot(aes(x = Workout.Number,
             y = `Predicted.2k.new`,
             color = Name,
             group = Name)) +
  geom_line() +
  geom_point() +
  labs(
    x = "Workout Number",
    y = "Predicted 2k",
    title = "Predicted 2k's"
```

```
  )
```




This is a small subset of rowers from this year. What would you say about this chart? Is there another chart that could work better here?

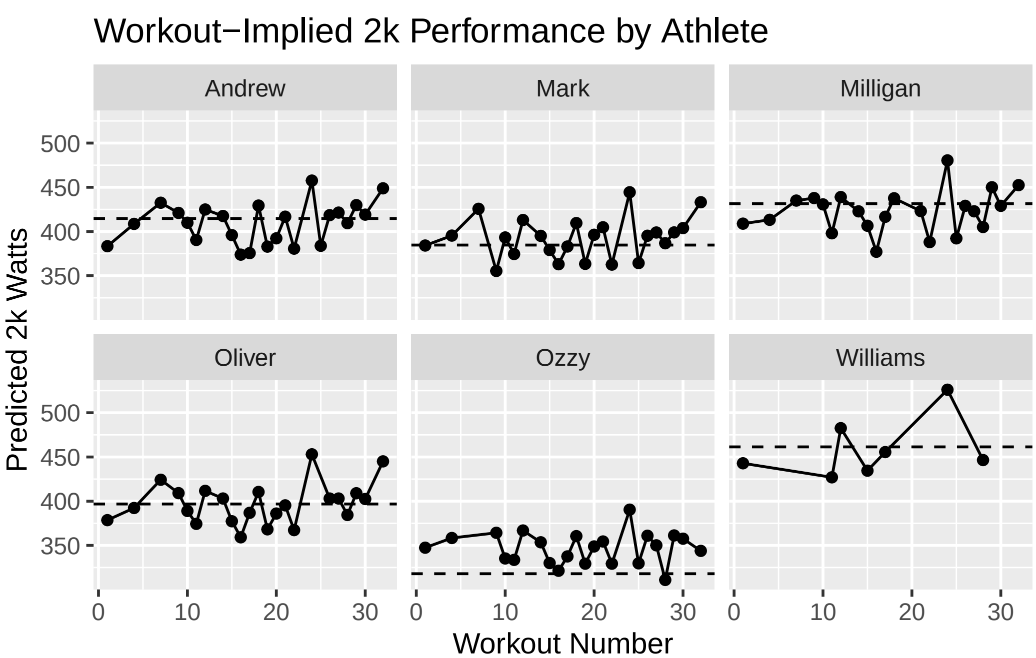
```
Calibration =  data.no.hr|>
    filter(Workout.Number == 31) |>
    dplyr::select(`2k.Watts`, `Predicted.2k.new`) |>
    mutate(delta = `2k.Watts`-`Predicted.2k.new`)

summary(
    Calibration,
    mean = mean()
) |>

knitr::kable()
```

| 2k.Watts | Predicted.2k.new | delta |
|---|---|---|
| Min. :307.6 | Min. :309.5 | Min. :-1.881 |
| 1st Qu.:352.7 | 1st Qu.:349.2 | 1st Qu.: 3.578 |
| Median :378.8 | Median :371.8 | Median : 6.996 |
| Mean :378.9 | Mean :371.6 | Mean : 7.272 |
| 3rd Qu.:405.8 | 3rd Qu.:395.0 | 3rd Qu.:10.750 |
| Max. :461.4 | Max. :442.4 | Max. :19.016 |

This is a numerical summary that helps us judge the accuracy of our model. How would you go about this procedure?

```
library(dplyr)
library(ggplot2)
library(stringr)


athletes <- c("Mark", "Williams", "Milligan", "Andrew", "Oliver","Ozzy")

data.no.hr |>
```

```
  mutate(Name = str_extract(Name, "^[A-Za-z]+")) |>
  filter(
    Workout.Number != 31,
    !is.na(Predicted.2k.new),
    Name %in% athletes
  ) |>
  ggplot(aes(x = Workout.Number, y = Predicted.2k.new)) +
  geom_point() +
  geom_line() +
  geom_hline(aes(yintercept = `2k.Watts`), linetype = "dashed") +
  facet_wrap(~Name) +
  labs(
    x = "Workout Number",
    y = "Predicted 2k Watts",
    title = "Workout-Implied 2k Performance by Athlete"
  )
```




```
library(dplyr)
library(stringr)
library(broom)
library(knitr)

athletes <- c("Mark", "Williams", "Milligan", "Andrew", "Oliver","Ozzy")

table_out <- data.no.hr |>
  mutate(Name = str_extract(Name, "^[A-Za-z]+")) |>
  filter(Name %in% athletes, !is.na(Predicted.2k.new)) |>
  group_by(Name) |>
  do({
    m <- lm(Predicted.2k.time.s ~ Workout.Number, data = .)
    tibble(
      Intercept = coef(m)[1],
      Slope = coef(m)[2],
      R2 = summary(m)$r.squared,
      p_value = summary(m)$coefficients[2,4]
```

```
    )
  }) |>
  ungroup()

kable(
  table_out,
  digits = 3,
  col.names = c("Athlete","Intercept (sec)","Slope (sec/workout)","R²","p-value"),
  caption = "Linear Model of Predicted 2k Time vs Workout Number"
)
```

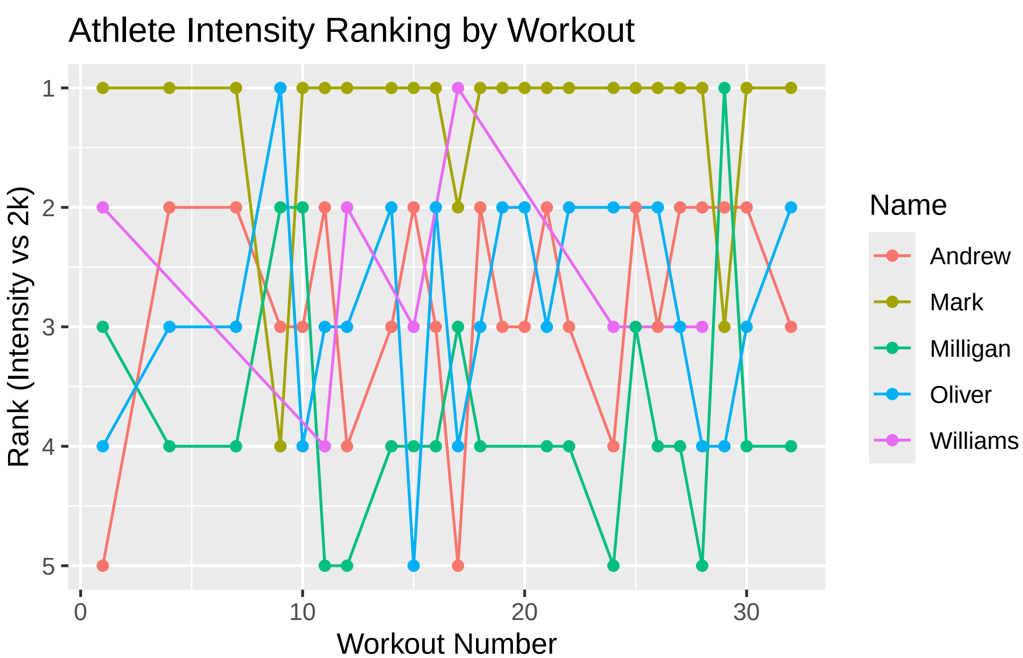
Table 2: Linear Model of Predicted 2k Time vs Workout Number

| Athlete | Intercept (sec) | Slope (sec/workout) | $R^2$ | p-value |
|---|---|---|---|---|
| Andrew | 383.652 | -0.196 | 0.061 | 0.234 |
| Mark | 387.961 | -0.135 | 0.026 | 0.441 |
| Milligan | 378.399 | -0.130 | 0.030 | 0.426 |
| Oliver | 387.439 | -0.191 | 0.055 | 0.270 |
| Ozzy | 400.004 | 0.101 | 0.015 | 0.573 |
| Williams | 369.365 | -0.179 | 0.047 | 0.608 |

This is a little bit outside the scope of our course. If you wanted to answer the question: are athletes getting faster from the start of the season to the end, what would you do? (hint: hypothesis testing)

```
library(dplyr)

# attach true 2k watts (Workout 31) to each athlete
data.model <- data.no.hr |>
  group_by(Name) |>
  mutate(`2k.Watts` = Watts[Workout.Number == 31][1]) |>
  ungroup()

# fit global fatigue exponent
fit <- nls(
  `2k.Watts` ~ Watts * (Time..s. / 390)^d,
  data = data.model |> filter(Workout.Number != 31),
  start = list(d = 0.07)
)

# extract fitted exponent
d_hat <- coef(fit)["d"]

# compute predicted 2k watts for every workout
data.model <- data.model |>
  mutate(
    Predicted.2k = Watts * (Time..s. / 390)^d_hat
  )

# optional: check prediction stability by athlete
sd_table <- data.model |>
  group_by(Name) |>
  summarize(
    sd_pred = sd(Predicted.2k.new, na.rm = TRUE),
    mean_pred = mean(Predicted.2k.new, na.rm = TRUE)
  )
```

```r
athletes <- c("Mark", "Williams", "Milligan", "Andrew", "Oliver")

data.rank <- data.model |>
  mutate(Relative.Intensity = Watts / `2k.Watts`)|>
  filter(Name %in% athletes) |>
  group_by(Workout.Number) |>
  mutate(Rank = rank(-Relative.Intensity)) |>
  ungroup()

library(ggplot2)

data.rank |>
  filter(Workout.Number != 31) |>
  ggplot(aes(x = Workout.Number, y = Rank, color = Name, group = Name)) +
  geom_line() +
  geom_point() +
  scale_y_reverse() +
  labs(
    x = "Workout Number",
    y = "Rank (Intensity vs 2k)",
    title = "Athlete Intensity Ranking by Workout"
  )
```




This graphic was the one that my team and coaching staff looked at the most. Why would I include it? What actionable insights can be derived from it? How would you change it, or add to it?